

Your University
Your Department

Thesis defense

Your Thesis Title

Author: Your Name

Advisor: Prof. Your Advisor

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Goals and objectives

Goal: improve the effectiveness of the proposed approach in the target domain

Objectives:

- review existing approaches and identify limitations
- analyze requirements and constraints
- design and implement a prototype system
- evaluate through experimental research
- compare results with existing alternatives

Input and output data

Input data:

- dataset of sample measurements
- configuration parameters
- baseline metrics from existing solutions

Output data:

- prototype implementation
- experimental results
- comparative analysis report

System analysis results

Functions:

- data ingestion and preprocessing
- analysis and transformation
- result generation and visualization
- logging and monitoring
- reporting and export

Subsystems:

- data processing subsystem
- analysis engine subsystem
- user interface subsystem
- monitoring and logging subsystem

Design evaluation

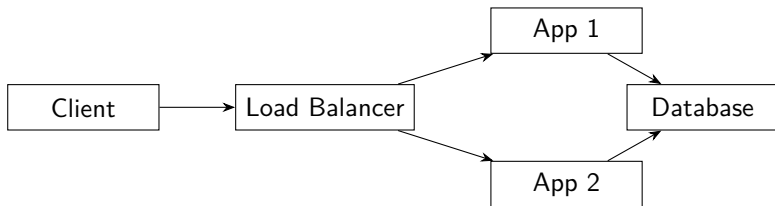
Goal: select the optimal architecture

Criteria: performance, accuracy, scalability, maintainability

Alternative	Score
Monolithic	7.15
Microservices	7.95
Hybrid	8.30

Selected approach: **Hybrid architecture**

System architecture



Experimental results

Performance comparison

Method	Small	Medium	Large
Baseline A	0.8 s	45.2 s	512.0 s
Baseline B	0.5 s	38.7 s	478.3 s
Proposed	0.3 s	22.1 s	287.5 s

37–44% improvement over baselines

Results

- comprehensive literature review completed
- system analysis with 5 functions and 4 subsystems
- comparative design evaluation (hybrid approach selected)
- prototype implemented in Python + PostgreSQL + Docker
- 37–44% performance improvement demonstrated
- 94.2% accuracy achieved

Conclusion

- analyzed existing approaches and identified key limitations
- designed modular architecture with clear subsystem boundaries
- implemented working prototype with comprehensive test coverage
- demonstrated significant improvements in performance and accuracy
- proposed directions for future work